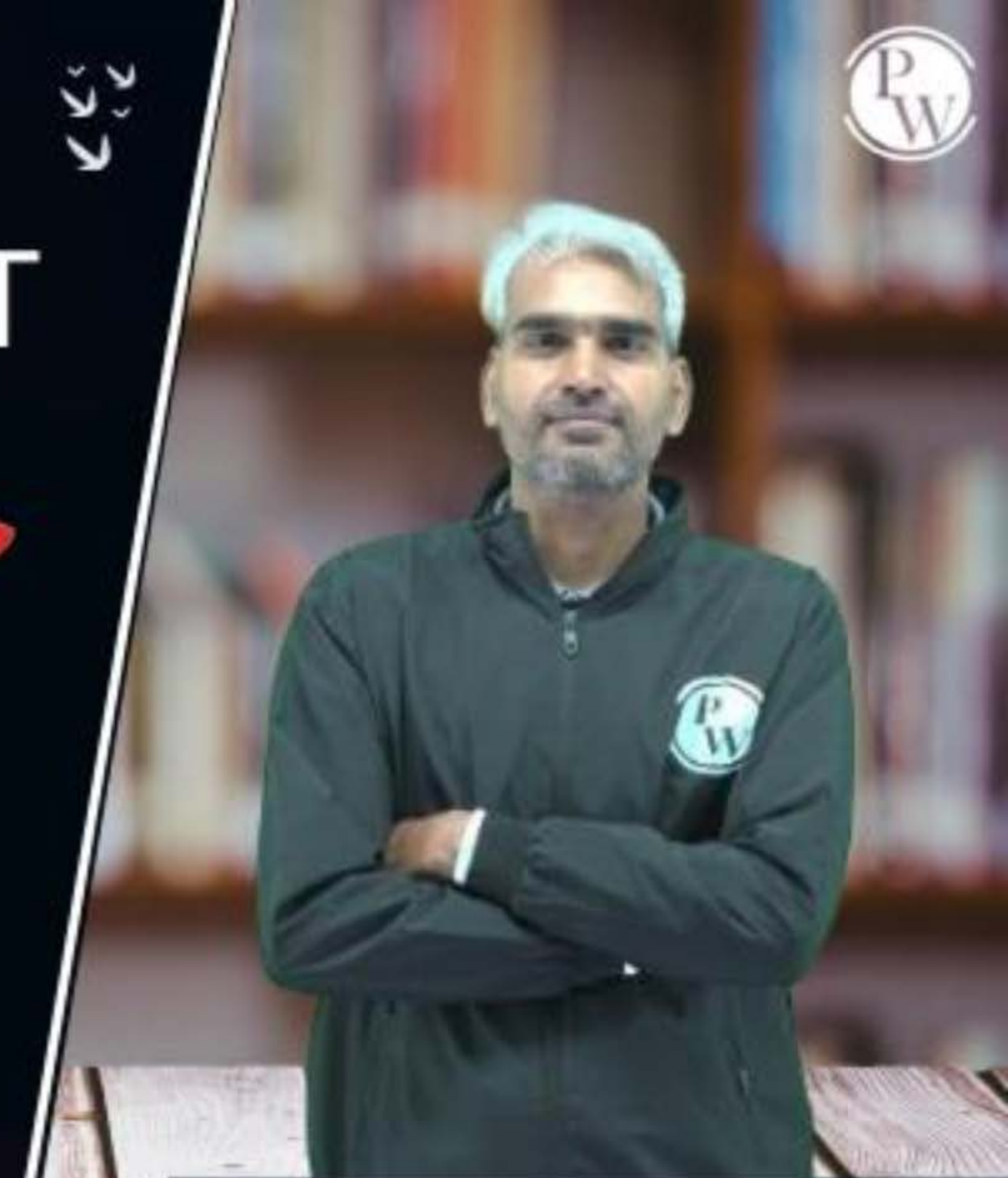
Computer Science & IT

Discrete Mathematics

Combinatorics

Lecture No. 06



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Recap of Previous Lecture

Topic Solution of recurrence relation using method of characteristic roots

Topic Generating functions

Topics to be Covered

Topic Generating functions

Topic Practice questions

Slide

Topic : Generating function

- Generating function is a way of generating an infinite sequence of numbers (a_n) by treating them as the coefficients of a formal power series.

$$a_0, \ a_1, \ a_2, \ a_3, \ \ldots\ldots$$

Generating function for given sequence $GF(x)$

$$= \underbrace{a_0 \cdot x^0 + a_1 \cdot x^1 + a_2 \cdot x^2 + a_3 \cdot x^3 + \ldots}_{\text{Solution of this series is called generating function in closed form}}$$

eg:- Consider the following sequence of real numbers.

1, 1, 1, 1, ...

Find generating function for the above sequence in Closed form.

Solu^n:

$$\begin{array}{lccccc} & a_0 & a_1 & a_2 & a_3 & a_4 \\ \text{Sequence} = & 1, & 1, & 1, & 1, & 1, \ldots \end{array}$$

Generating function $Gf(x)$ (in the form of Power series) $= 1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 + 1 \cdot x^3 + \ldots$

$$= \underbrace{1 + x + x^2 + x^3 + \ldots\ldots}$$

it is an infinite GP with Common ratio 'x'

$\left\{ \text{Summation of infinite GP} = \dfrac{a}{1-r} \right\}$

$$Gf(x) = \frac{1}{1-x}$$

first term

Common ratio

$$\boxed{Gf(x) = \frac{1}{(1-x)}}$$

Consider the following sequence of real numbers.

$$a_0, a_1, a_2, a_3, \ldots$$

Generating function

$$G(x) = a_0 x^0 + a_1 \cdot x^1 + a_2 \cdot x^2 + a_3 \cdot x^3 + \ldots$$

$$\boxed{Gf(x) = \sum_{i=0}^{\infty} (a_i) x^i}$$

a_i is called general term

Binomial Expansion:-

$$(a+b)^n = {}^nC_0 (a)^n \cdot (b)^0 + {}^nC_1 (a)^{n-1} \cdot (b)^1 + {}^nC_2 (a)^{n-2} \cdot (b)^2 + \cdots + {}^nC_r (a)^{n-r} \cdot (b)^r + \cdots + {}^nC_n (a)^0 (b)^n$$

$$= {}^nC_0 \cdot a^0 \cdot b^n + {}^nC_1 \cdot a^1 \cdot (b)^{n-1} + {}^nC_2 \cdot a^2 \cdot b^{n-2} + \cdots + {}^nC_r (a)^r \cdot (b)^{n-r} + \cdots + {}^nC_n (a)^n \cdot (b)^0$$

$$(1+x)^n = {}^nC_0 (1)^n \cdot x^0 + {}^nC_1 (1)^{n-1} \cdot (x)^1 + {}^nC_2 (1)^{n-2} (x)^2 + \cdots + {}^nC_r (1)^{n-r} (x)^r + \cdots + {}^nC_n (1)^0 x^n$$

$$\boxed{(1+x)^n = {}^nC_0 \cdot x^0 + {}^nC_1 \cdot x^1 + {}^nC_2 \cdot x^2 + \cdots + {}^nC_r \cdot x^r + \cdots + {}^nC_n \cdot x^n}$$

$$\boxed{(1+x)^n = \sum_{i=0}^{n} \left({}^nC_i\right) x^i}$$

Q: Consider the following seq. of real numbers.

$$^nC_0, ^nC_1, ^nC_2, ^nC_3, \ldots, ^nC_n$$

then what will be the generating function

Solu: $Gf(x) = {}^nC_0 \cdot x^0 + {}^nC_1 \cdot x^1 + {}^nC_2 \cdot x^2 + \cdots + {}^nC_n \cdot x^n$

$$\boxed{Gf(x) = (1+x)^n}$$

eg:- Consider the following sequence of real numbers.

$$\overset{a_0}{1}, \overset{a_1}{2}, \overset{a_2}{3}, \overset{a_3}{4}, \overset{a_4}{5}, \ldots\ldots \quad \{\text{it is a sequence of natural no:s}\}$$

Find the generating function for the above sequence

Generating function $Gf(x) = 1\cdot x^0 + 2\cdot x^1 + 3\cdot x^2 + 4\cdot x^3 + \ldots\ldots + (r+1)\cdot x^r + \ldots\ldots$

$$\boxed{\frac{d}{dx}(x^n) = n \cdot x^{(n-1)}}$$

$$\frac{1}{1-x} = 1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 + 1 \cdot x^3 + \cdots + 1(x)^{r+1} + \cdots$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx}\left\{1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 + 1 \cdot x^3 + \cdots + 1(x)^{r+1} + \cdots\right\}$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \left\{0 + 1 + 2 \cdot x + 3 \cdot x^2 + \cdots + (r+1) \cdot x^r + \cdots\right\}$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = 1 \cdot x^0 + 2 \cdot x^1 + 3 \cdot x^2 + \cdots + (r+1) \cdot x^r + \cdots$$

= Gregenating function for sequence 1, 2, 3, 4, - - - .

eg:- Consider the following sequence of real numbers.

$$\begin{matrix} a_0 & a_1 & a_2 & a_3 & a_4 & \\ 1, & 2, & 3, & 4, & 5, & \ldots\ldots \end{matrix}$$

Find the generating function for the above sequence

Generating function $Gf(x) = 1\cdot x^0 + 2\cdot x^1 + 3\cdot x^2 + 4\cdot x^3 + \ldots\ldots + (r+1)\cdot x^r + \ldots\ldots$

$$Gf(x) = \frac{d}{dx}\left(\frac{1}{1-x}\right)$$

$$\boxed{\frac{d}{dx}\left(\frac{P}{Q}\right) = \frac{P'Q - PQ'}{Q^2}}$$

$$= \frac{\left(\frac{d}{dx}(1)\right)(1-x) - 1\cdot\frac{d}{dx}(1-x)}{(1-x)^2} = \frac{0 - 1\cdot(0-1)}{(1-x)^2}$$

$$\boxed{Gf(x) = \frac{1}{(1-x)^2}}$$

eg:- Find the Closed form of generating function.

$$Gf(x) = 0\cdot x^0 + 1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \cdots$$

Sequence = 0, 1, 2, 3, ... { it is a sequence of whole numbers }

Solu^n.

$$Gf(x) = 0\cdot x^0 + 1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \cdots + r\cdot(x)^r + \cdots$$

We know, $\frac{1}{(1-x)^2} = 1\cdot x^0 + 2\cdot x^1 + 3\cdot x^2 + \cdots + (r+1)\cdot x^r + \cdots$

multiply L.H.S & R.H.S by x

$$\frac{x}{(1-x)^2} = 1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \cdots + (r+1)\cdot x^{(r+1)} + \cdots$$

if we multiply it by 'x' then it becomes $(r+1)\cdot x^{r+1}$

$$\boxed{\frac{x}{(1-x)^2} = 0\cdot x^0 + 1\cdot x^1 + 2\cdot x^2 + 3\cdot x^3 + \cdots + (r+1)\cdot x^{(r+1)} + \cdots}$$

∴ Generating function w.r.t. 0, 1, 2, 3, 4, ... is $\frac{x}{(1-x)^2}$

* If we want to right shift a sequence by 'K' position then multiply the sequence by x^K.

$\therefore$ The generating function for sequence

$$\underbrace{0, 0, 0}_{\substack{\text{Right shift} \\ \text{by '3' position} \\ \therefore \text{ multiply by } x^3}}, \underbrace{1, 1, 1, 1, 1, 1, 1, \ldots\ldots}_{\substack{\text{generating function is} \\ \left(\frac{1}{1-x}\right)}} \text{ will be}$$

Hence generating fn for given seq. will be $x^3 \cdot \left(\frac{1}{1-x}\right) = \left(\frac{x^3}{1-x}\right)$

Binomial Coefficient

We know, $${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!} = \frac{n\cdot(n-1)\cdot(n-2)\cdot\ldots\cdot(n-r+1)\cdot\overbrace{(n-r)(n-r-1)\ldots3\cdot2\cdot1}^{\cancel{(n-r)!}}}{(r\cdot(r-1)(r-2)\ldots3\cdot2\cdot1)\cdot\cancel{(n-r)!}}$$

$$\boxed{{}^{n}C_{r} = \frac{(n-0)\cdot(n-1)\cdot(n-2)\cdot\ldots\cdot(n-(r-1))}{r!}}$$

from 0 to $(r-1)$ we have 'r' terms $\left\{\begin{array}{l}\text{we perform subtraction} \\ \text{from 0 to } (r-1)\end{array}\right\}$

Binomial Coefficient

from '0' 'to' $(r-1)=3$

1

$${}^{6}C_{4} = \frac{(6-0)\cdot(6-1)\cdot(6-2)\cdot(6-3)}{4!}$$

$r=4$,

$\therefore (r-1)=(4-1)=3$

$$= \frac{\overset{3}{\cancel{6}} * 5 * \cancel{4} * \cancel{3}}{\cancel{4} * \cancel{3} * \cancel{2} * 1} = 15 \text{ Ans}$$

Extended Binomial Coefficient

Find the value of ${}^{-5}C_3$

$$ {}^{-5}C_3 = \frac{(-5-0)\,(-5-1)\cdot(-5-2)}{3!} = \frac{-5 * -\cancel{6} * -7}{\cancel{6}} $$

$r = 3$

$(r-1) = (3-1) = 2$

$\therefore$ subtract from '0' to '2'

$$ = (-5)\cdot(-1)\cdot(-7) $$

$$ = -35 $$

Ans

Extended Binomial Coefficient

We Know $^nC_r = \dfrac{n!}{r!\,(n-r)!}$

Find the value of $^{-n}C_r$ { where 'n' & 'r' are non-negative integers }

$$\boxed{^{-n}C_r = (-1)^r \cdot {}^{(n+r-1)}C_r}$$

$$^{-n}C_r = \frac{(-n-0)\cdot(-n-1)\cdot(-n-2)\cdot\ldots\ldots(-n-(r-1))}{r!}$$

Subtract from 0 to $(r-1)$

$$= (-1)^r \cdot \frac{(n+0)\cdot(n+1)\cdot(n+2)\cdot\ldots\ldots(n+(r-1))}{r!}$$

$$= (-1)^r \cdot \frac{(n+r-1)\cdot(n+r-2)\cdot\ldots\ldots(n+2)\cdot(n+1)\cdot n}{r!} * \frac{(n-1)\cdot(n-2)\cdot\ldots\ldots 3\cdot 2\cdot 1}{(n-1)\cdot(n-2)\cdot\ldots\ldots 3\cdot 2\cdot 1}$$

$$= (-1)^r \cdot \frac{(n+r-1)!}{r!\,(n-1)!} = (-1)^r \cdot \frac{(n+r-1)!}{r!\,((n+r-1)-r)!} = (-1)^r \cdot {}^{(n+r-1)}C_r$$

$(n+r-1)-r = (n-1)$

$$\boxed{^{-n}C_r = (-1)^r \cdot {}^{(n+r-1)}C_r}$$

Topic : Extended binomial coefficient

$$\boxed{{}^{-n}C_r = (-1)^r \cdot {}^{(n+r-1)}C_r}$$

Where 'n' & 'r' are non-negative integers.

Q:- ⇒ Find the Coefficient of x^r w.r.t. following generating functions.

① $Gf(x) = (1+x)^n = {}^nC_0 \cdot x^0 + {}^nC_1 \cdot x^1 + {}^nC_2 \cdot x^2 + \cdots + {}^nC_r \cdot x^r + \cdots$

$$Gf(x) = \sum_{i=0}^{\infty} \underbrace{\left({}^nC_i\right)}_{\text{it is the general term}} \cdot x^i$$

it is the general term {it gives the coefficient of x^i}

Coefficient of $x^r = {}^nC_r$

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② $Gf(x) = (1-x)^{-n}$

$= (1+(-x))^{-n} = {}^{-n}C_0 \cdot (-x)^0 + {}^{-n}C_1 \cdot (-x)^1 + {}^{-n}C_2 \cdot (-x)^2 + \cdots + {}^{-n}C_r \cdot (-x)^r + \cdots$

$$Gf(x) = \sum_{i=0}^{\infty} \underbrace{\left({}^{(n+i-1)}C_i\right)}_{\text{general term}} \cdot x^i$$

general term {Coefficient of x^i}

Coefficient of $x^r = {}^{(n+r-1)}C_r$

${}^{-n}C_r \cdot (-x)^r = \{{}^{-n}C_r\}\{(-1)^r \cdot (x)^r\}$

$= \{(-1)^r \cdot {}^{n+r-1}C_r\} \cdot \{(-1)^r \cdot x^r\}$

$= \cancel{(-1)^{2r}} \cdot {}^{(n+r-1)}C_r \cdot x^r$

$= 1 \cdot {}^{(n+r-1)}C_r \cdot x^r$

H.W. 3 $Gf(x) = (1-x)^n =$

$$= (1+(-x))^n = {}^nC_0(-x)^0 + {}^nC_1\cdot(-x)^1 + {}^nC_2\cdot(-x)^2 + \cdots + {}^nC_r(-x)^r + \cdots + {}^nC_n(-x)^n$$

$$Gf(x) = \sum_{r=0}^{n} \left((-1)^r \cdot {}^nC_r\right) x^r$$

general term

$${}^nC_r\cdot(-1)^r\cdot(x)^r = (-1)^r\cdot {}^nC_r \cdot x^r$$

$\boxed{\text{Coefficient of } x^r = (-1)^r\cdot {}^nC_r}$

H.W. 4 $Gf(x) = (1+x)^{-n} = {}^{-n}C_0(x)^0 + {}^{-n}C_1\cdot(x)^1 + {}^{-n}C_2\cdot(x)^2 + \cdots + {}^{-n}C_r\cdot(x)^r + \cdots$

$$Gf(x) = \sum_{r=0}^{\infty} \left((-1)^r\cdot {}^{(n+r-1)}C_r\right) x^r$$

general term

$${}^{-n}C_r\cdot(x)^r \equiv (-1)^r\cdot {}^{(n+r-1)}C_r\cdot(x)^r$$

$$\left((-1)^r\cdot {}^{(n+r-1)}C_r\right)(x)^r$$

Coefficient of $x^r = (-1)^r\cdot {}^{(n+r-1)}C_r$

H.W.

⑤ $Gf(x) = (1+ax)^n$

$= (1+(ax))^n = {}^nC_0 (ax)^0 + {}^nC_1 (ax)^1 + {}^nC_2 (ax)^2 + \dots + {}^nC_r (ax)^r + \dots + {}^nC_n (ax)^n$

$$Gf(x) = \sum_{r=0}^{n} \underbrace{(a^r \cdot {}^nC_r)}_{\text{general term}} \cdot x^r$$

${}^nC_r (ax)^r \equiv {}^nC_r \cdot (a)^r \cdot (x)^r \equiv (a)^r \cdot {}^nC_r \cdot (x)^r$

Coefficient of $x^r = (a)^r \cdot {}^nC_r$

H.W.

⑥ $Gf(x) = (1-ax)^{-n}$

$= (1+(-ax))^{-n} = {}^{-n}C_0 (-ax)^0 + {}^{-n}C_1 (-ax)^1 + {}^{-n}C_2 (-ax)^2 + \dots + {}^{-n}C_r (-ax)^r + \dots$

$$Gf(x) = \sum_{r=0}^{\infty} \underbrace{\left((a)^r \cdot {}^{(n+r-1)}C_r\right)}_{\text{general term}} (x)^r$$

${}^{-n}C_r = (-1)^r \cdot {}^{(n+r-1)}C_r$, $(-ax)^r = (-a)^r (x)^r$

$(-1)^r \cdot {}^{(n+r-1)}C_r \cdot (-1)^r (a)^r (x)^r$

$= (a)^r \cdot {}^{(n+r-1)}C_r \cdot (x)^r$

Coefficient of $x^r = (a)^r \cdot {}^{(n+r-1)}C_r$

H.W

7) $Gf(x) = (1-ax)^n \equiv {}^nC_0(-ax)^0 + {}^nC_1(-ax)^1 + {}^nC_2(-ax)^2 + \dots + {}^nC_r(-ax)^r + \dots + {}^nC_n(-ax)^n$

$${}^nC_r(-ax)^r \equiv {}^nC_r \cdot (-a)^r \cdot (x)^r \equiv \underbrace{(-1)^r (a)^r \cdot {}^nC_r \cdot (x)^r}$$

$$Gf(x) = \sum_{i=0}^{n} \left((-1)^i \cdot (a)^i \cdot {}^nC_i\right)(x)^i$$

general term

Coefficient of $x^r = (-1)^r \cdot (a)^r \cdot {}^nC_r$

H.W

8) $Gf(x) = (1+ax)^{-n} = {}^{-n}C_0 \cdot (ax)^0 + {}^{-n}C_1 \cdot (ax)^1 + {}^{-n}C_2 (ax)^2 + \dots + {}^{-n}C_r \cdot (ax)^r + \dots$

$${}^{-n}C_r = (-1)^r \cdot {}^{(n+r-1)}C_r \qquad (ax)^r = (a)^r \cdot (x)^r$$

$$(-1)^r \cdot (a)^r \cdot {}^{(n+r-1)}C_r \cdot (x)^r$$

$$Gf(x) = \sum_{i=0}^{\infty} \left((-1)^i \cdot (a)^i \cdot {}^{(n+i-1)}C_i\right)(x)^i$$

general term

Coefficient of $x^r = (-1)^r (a)^r \cdot {}^{(n+r-1)}C_r$

Gate-PYQ

We know $\frac{1}{(1-x)^2}$ is the generating function for sequence $\underset{a_0}{1}, \underset{a_1}{2}, \underset{a_2}{3}, \underset{a_3}{4}, \dots \underset{a_i}{(i+1)}$

$\therefore g(i) = (i+1)$

#Q. Let $G(x) = \frac{1}{(1-x)^2} = \sum_{i=0}^{\infty} g(i)\, x^i$, where $|x| < 1$. What is $g(i)$?

it is coefficient of $(x)^i$

$$\frac{1}{(1-x)^2} = (1-x)^{-2} \Rightarrow$$

Coefficient of $(x)^i$ in the expansion of $(1-x)^{-n} = {}^{(n+i-1)}C_i$

$\therefore$ Coefficient of $(x)^i$ in the expansion of $(1-x)^{-2} = {}^{(2+i-1)}C_i = {}^{(i+1)}C_i = {}^{(i+1)}C_{(i+1)-i} = {}^{(i+1)}C_1 = (i+1)$

${}^nC_r = {}^nC_{n-r}$

$\therefore g(i) = (i+1)$

#Q. If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to?

$\frac{(1+z)}{(1-z)^3}$ is the generating function for sequence $a_0, a_1, a_2, a_3, a_4 \ldots$

then $\frac{(1+z)}{(1-z)^3} = a_0 \cdot z^0 + a_1 \cdot z^1 + a_2 \cdot z^2 + a_3 \cdot z^3 + \ldots$

Coefficient of z^0

Coefficient of z^3

$$\frac{(1+z)}{(1-z)^3} = (1+z) \cdot (1-z)^{-3} = (1 \cdot z^0 + 1 \cdot z^1) \cdot (1-z)^{-3}$$

'1' is the coefficient of $(z)^1$.

'1' is the Coefficient of z^0 in the expansion of $(1+z)$

Overall Coefficient of z^0 in the Expansion of $\frac{(1+z)}{(1-z)^3}$ $= \left\{\begin{matrix}\text{Coefficient of } z^0 \\ \text{in the expansion} \\ \text{of } (1+z)\end{matrix}\right\} \cdot \left\{\begin{matrix}\text{Coefficient of } z^0 \\ \text{in the expansion} \\ \text{of } (1-z)^{-3}\end{matrix}\right\}$.

Overall Coefficient of z^3 in the Expansion of $\frac{(1+z)}{(1-z)^3}$ $= \left[\left(\left\{\begin{matrix}\text{Coefficient of } z^0 \\ \text{in the expansion} \\ \text{of } (1+z)\end{matrix}\right\} \cdot \left\{\begin{matrix}\text{Coefficient of } z^3 \\ \text{in the expansion} \\ \text{of } (1-z)^{-3}\end{matrix}\right\}\right) + \left(\left\{\begin{matrix}\text{Coefficient of } z^1 \\ \text{in the expansion} \\ \text{of } (1+z)\end{matrix}\right\} \cdot \left\{\begin{matrix}\text{Coefficient of } z^2 \\ \text{in the expansion} \\ \text{of } (1-z)^{-3}\end{matrix}\right\}\right)\right]$

Overall Coefficient of z^0 in the Expansion of $\frac{(1+z)}{(1-z)^3}$ $= \begin{Bmatrix}\text{Coefficient of } z^0 \\ \text{in the expansion} \\ \text{of } (1+z)\end{Bmatrix} \cdot \begin{Bmatrix}\text{Coefficient of } z^0 \\ \text{in the expansion} \\ \text{of } (1-z)^{-3}\end{Bmatrix}$

$$= 1 \cdot {}^{3+0-1}C_0 = 1 \times {}^2C_0 = 1 \times 1 = 1$$

$$\therefore \boxed{a_0 = 1}$$

Overall Coefficient of z^3 in the Expansion of $\frac{(1+z)}{(1-z)^3}$ $= \left[\left(\begin{Bmatrix}\text{Coefficient of } z^0 \\ \text{in the expansion} \\ \text{of } (1+z)\end{Bmatrix} \cdot \begin{Bmatrix}\text{Coefficient of } z^3 \\ \text{in the expansion} \\ \text{of } (1-z)^{-3}\end{Bmatrix}\right) + \left(\begin{Bmatrix}\text{Coefficient of } z^1 \\ \text{in the expansion} \\ \text{of } (1+z)\end{Bmatrix} \cdot \begin{Bmatrix}\text{Coefficient of } z^2 \\ \text{in the expansion} \\ \text{of } (1-z)^{-3}\end{Bmatrix}\right)\right]$

$$\left(1 \times {}^{3+3-1}C_3\right) + \left(1 \times {}^{3+2-1}C_2\right)$$

$$(1 \times 10) + (1 \times 6)$$

$$\boxed{a_3 = 10 + 6 = 16}$$

#Q. If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to?

$$a_3 - a_0 = \underbrace{16}_{\text{Coefficient of } z^3} - \underbrace{1}_{\text{Coefficient of } z^0} = 15 \quad \underline{\text{Ans}}$$

Q:- Find the Coefficien of x^{12} in the expansion of

$$\left(\underbrace{x^3+x^4+x^5+x^6+\cdots\cdots}_{\text{It is an infinite GP} \Rightarrow \therefore \text{ apply Summation of GP} = \left(\frac{a}{1-r}\right)}\right)^3$$

'a'

$$= \left(\frac{x^3}{(1-x)}\right)^3 = \left(x^3 \cdot (1-x)^{-1}\right)^3$$

'r'

$$= \underbrace{x^9}_{} \cdot \underbrace{(1-x)^{-3}}_{}$$

$$\left(\begin{matrix}\text{Coefficient} \\ \text{of } x^9\end{matrix} * \begin{matrix}\text{Coefficient} \\ \text{of } x^3\end{matrix}\right) = \begin{matrix}\text{Coefficient} \\ \text{of } x^{12}\end{matrix}$$

$$= 1 * \left({}^{3+3-1}C_3\right)$$

$$= 1 * {}^5C_3 = 1 * 10 = 10 \text{ Ans}$$

General term is given, and we need to identify the generating function.

#Q. Which one the following is a closed from expression for the generating function of the sequence$\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0,1,2, \ldots$?

$$Gf(x) = \sum_{n=0}^{\infty} (a_n) \cdot x^n$$

$$= \sum_{n=0}^{\infty} (2n+3) \cdot x^n$$

A) $\frac{3}{(1-x)^2}$

B) $\frac{3x}{(1-x)^2}$

C) $\frac{2-x}{(1-x)^2}$

D) $\frac{3-x}{(1-x)^2}$

$$Gf(x) = \sum_{n=0}^{\infty} (2n+3)\cdot x^n$$

$$= \sum_{n=0}^{\infty} 2n\cdot(x)^n + \sum_{n=0}^{\infty} 3\cdot(x)^n$$

$$= \left(2\cdot\underbrace{\sum_{n=0}^{\infty} n\cdot(x)^n}\right) + \left(3\cdot\underbrace{\sum_{n=0}^{\infty} (1)\cdot(x)^n}\right)$$

general term $= n$

$\therefore$ Seq. : $0, 1, 2, 3, \ldots$

We know $gf : \dfrac{x}{(1-x)^2}$

general term $= 1$ $\therefore$ Seq $= 1, 1, 1, \ldots$

Hence $gf : \dfrac{1}{(1-x)}$

$$\left(2\cdot\frac{x}{(1-x)^2}\right) + \left(3\cdot\frac{1}{(1-x)}\right)$$

$$= \frac{2x}{(1-x)^2} + \frac{3}{(1-x)} = \frac{2x+3(1-x)}{(1-x)^2} = \frac{3-x}{(1-x)^2}$$

Ans

2 mins Summary

Topic Generating functions

Topic Practice questions

Slide

THANK - YOU

Slide